

CALIFORNIA'S GREEN RATING SYSTEM

Build It Green has developed an effective green rating system for the state of California.

BY KATY HOLLBACHER AND ELKA KARL

Although GreenPoint Rated has only recently made its public debut, it's hardly a new kid on the block. In fact, GreenPoint Rated is an outgrowth of successful green building programs and resources that have been serving Californians since 2000. Its origins lie in the green building guidelines first developed by Green Building in Alameda County. It is now managed by Build It Green to serve the entire state.

The guidelines, and the rating and third-party verification system that grew out of them, were developed and repeatedly refined by a group of production builders, contractors, architects and designers, multifamily home developers, state and local government leaders, regional and national building science experts, product manufacturers and suppliers, and green building advocates. In the past six years, numerous local governments have adopted and use the guidelines, and many encourage or require third-party rating.

The measures recommended in the guidelines, and the performance benchmarks in GreenPoint Rated, were specifically developed to address climate and market conditions in California, are endorsed by third-party sources, and are backed by sound building science. Builders and home buyers can feel confident that these measures and benchmarks have been

tried and tested in the field, and have been proven effective. And the process for updating the program over time is controlled by California decision makers and stakeholders, not by interests outside of the state.

Build It Green, which oversees GreenPoint Rated, is a professional nonprofit membership organization whose mission is to promote healthy, energy- and resource-efficient buildings in California. The GreenPoint Rated label gives home buyers assurance that they're getting a healthier, more environmentally responsible home. At the same time, builders, contractors, and architects/designers are provided with validation from a trusted, independent source to support their marketing efforts as green builders.

Other building professionals, such as real estate agents, financing specialists, appraisers, and others who work in fields related to new-home construction and sales, can use GreenPoint Rated to help their clients identify homes that are healthier and



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more environmentally friendly, and that exceed California's building and energy codes.

In addition, GreenPoint Rated fosters sustainability in communities by working with local governments to promote the development of healthier, more environmentally responsible homes. GreenPoint Rated also provides a means of tracking the number of green homes built in a given community, and of helping home owners and home builders to



Celebrating Sara Conner Court's GreenPoint Rated award are Hayward City Councilman Olden Henson, Eden Housing Board President Tim Reilly, and Kathy Wilson, district director for California state Senator Ellen Corbett.

understand the resource and health benefits those homes deliver.

An Easy Green Rating

To help home buyers and builders make progress on the path toward greener homes, GreenPoint Rated has been designed a bit like a report card. A GreenPoint Rated home is graded on five categories: Energy Efficiency, Resource Conservation, Indoor Air Quality, Water Conservation, and Community. If the home meets minimum point requirements in each category and scores more than 50 total points, it earns the right to bear the GreenPoint Rated label (see "GreenPoints are Changing").

But GreenPoint Rated goes beyond merely guaranteeing that the home exceeds California's minimum building and energy code requirements. It also provides a numerical score, which allows buyers to evaluate and compare the environmental performance of individual homes.

What's more, this scoring system gives builders an incentive for continuing to build greener as they gain experience with environmentally friendly products and construction practices. And it rewards builders whose homes have achieved higher levels of

performance than their competitors' homes.

Newly constructed single-family homes (custom and production) and multifamily homes in California are currently eligible for participation. GreenPoint Rated for remodeling for existing homes is under development.

GreenPoint Rated is a comprehensive program that specifically addresses the needs of California's residential building industry, but using it does not preclude participation in other green home rating systems. In fact, GreenPoint Rated was designed not to compete, but to be compatible, with programs such as Energy Star for Homes (which focuses on the energy efficiency of new homes); the U.S. Green Building Council's LEED for Homes program and the National Association of Home Builders guidelines (both national programs that are not tailored to specific regional conditions); and California Green Builder (a program developed by the building industry for production builders).

Depending on the builder's marketing needs, GreenPoint Rated can be used either independently or in conjunction with these other home rating systems. In some ways, GreenPoint Rated is seen as a stepping stone to other, more rigorous certification programs. For instance, while LEED has 18 major requirements, GreenPoint Rated has only 3. "We're a more economic option," says Bruce Mast, development director at Build It Green. He says that the program's affordability quotient encourages builders who are wary of more-expensive certification programs to develop green practices in their building.

"As the market has gotten softer, builders are trying to find a way to differentiate themselves from the competition," says Mast, when asked why builders choose to work with GreenPoint Rated. "Builders are finding that they have direct benefits in terms of direct sales, quicker sales, and higher market value."

GreenPoint's Latest Success Story

Last spring, during Affordable Housing Week, Sara Conner Court Apartments, located in Hayward, California, was honored with a GreenPoint Rated award. Sara Conner Court earned 72 points under the GreenPoint program, becoming the first affordable rental development in the San Francisco Bay Area's East Bay region to earn the GreenPoint Rated designation.

The Sara Conner project, which consists of 57 one-, two-, and three-bedroom units, is built on a 1.8-acre brownfield site previously occupied by a beverage processing plant, a gas station, and a dry cleaner. It was completed in August 2006 by owner/developer Eden Housing, Incorporated, with architect Pyatok Architects, of Oakland, and general contractor Segue Construction, of Point Richmond, California, working in partnership with the city of Hayward. Their goal was to create a community that was safe, attractive, environmentally friendly, and healthy for people with low incomes—and they succeeded. In addition to the GreenPoint Rated award, the project received a Bay Friendly Landscape award and a special recognition award from California state Senator Ellen Corbett.

"Eden Housing is committed to green building because it benefits our environment and our residents, by using water and energy more efficiently, by reducing waste, and by encouraging recycling," says Tim Reilly, president of the Eden Housing board of directors. Eden Housing plans to incorporate green building into other California housing projects, including

GreenPoints Are Changing

How hot water gets delivered to the various fixtures in a house has a lot to do with how much energy and water those fixtures use. That's why Build It Green, a nonprofit that promotes the construction of healthy and resource-efficient buildings, recently made some important changes to the plumbing section in its newest Green Buildings guidelines. Six categories of hot water distribution system now qualify for GreenPoints (see Table 1). GreenPoints are values assigned to recommended practices in the guidelines; the greater the benefits to the homeowner, and to the environment, the more strongly the practice is recommended and the more points it earns.

Improving the hot water distribution system reduces the volume of water in the pipes between the water heater and the fixtures. This water usually goes down the drain while the homeowner waits for hot water to arrive at the faucet. In a house with one central core plumbing system there are never more than four cups of water standing in the pipes between the water heater and any hot water fixture. Homes with this setup earn points in three categories: energy, resources, and water.

Some houses have what is known as a structured plumbing with recirculation loop system. These houses have a circulation loop located within 10 plumbing feet of each fixture and primed before each use with an on-demand pump. Houses with this setup get the second highest number of points.

This system allows for more widely separated hot water fixtures, and a wider variety of floor plans, than is possible with a central core plumbing system. (See "Benefits of Demand-Controlled Pumping," *HE* Sept/Oct '06 p. 18, to learn more about this type of setup.)

All six methods of distributing hot water are discussed in detail in both Build It Green's Home Remodeling and New Home Remodeling and New Home Remodeling guidelines, sections G. These and the Multifamily Residential Green Building guidelines offer suggestions for



Build It Green recently changed its GreenPoints plumbing section.

Table 1. Six Ways to Distribute Domestic Hot Water Efficiently

Hot Water Distribution System	Saves Energy	Saves Resources	Saves Water
Hot water pipes from water heater to kitchen insulated	X		X
All hot water pipes insulated	X		X
Engineered parallel piping			X
Engineered parallel piping with recirculation loop			X
Structured plumbing with recirculation loop	X		X
Central core plumbing	X	X	X

conserving natural resources, using water and energy wisely, improving indoor air quality, and planning for livable and vibrant communities. Builders who follow these guidelines create healthy, durable homes that are easier on the environment and cost less to operate and maintain.

—Gary Klein and Brian Gitt

Gary Klein is an energy specialist with the California Energy Commission. Brian Gitt is the executive director of Build It Green.



BUILD IT GREEN

Build It Green has been greening California homes since 2000.

those in San Leandro, Richmond, Novato, and Vallejo.

While Sara Conner Court doesn't have glamorous green features like solar electric systems or vegetated roofs, the team improved the project's environmental performance while maintaining its affordability by focusing on such fundamental strategies as solar orientation, daylighting, natural ventilation, and durability. Since lower energy use and lower greenhouse gas emissions are required by California's Title 24–2001, the apartments have no air conditioning; rather, they were designed for good natural ventilation, with operable windows at the front and back of each unit to provide cross breezes. The courtyard and breezeways ensure that each unit gets plenty of daylight and fresh air.

The apartments' efficient gas-fueled hydronic heating systems allow the water heater to do double duty, providing hot water to the faucets as well as heating the apartment. Each apartment has its own gas meter, which provides an incentive for the residents to keep their energy use—and related greenhouse gas emissions—in

check. All heating ducts were fully sealed to improve the efficiency of the heating system and protect indoor air quality (IAQ).

Eden was committed to exceeding the requirements of California's design process; however, the team realized that the largest of the four buildings was coming in somewhat below that energy performance benchmark. They analyzed extra measures that would improve the design's energy performance while still being affordable. The team opted for tighter sealing of the heating ducts. This saves energy and provides the added benefit of better IAQ. Also, even though the project was developed under Title 24–2001, Eden expected that the 2005 code update would require tighter duct sealing. They decided to stay ahead of the regulatory curve by choosing tighter duct sealing for this project.

Carpets in living rooms and bedrooms bear the Carpet and Rug Institute's Green Label, an indication of lower chemical emissions. In addition, the carpets were installed with adhesives that have low levels of VOCs. The kitchen flooring is natural linoleum instead of vinyl. The kitchen range hood's exhaust fans vent to the outside, and bathroom exhaust fans are on timers tied in to the light switch, so that they stay on long enough to vent the space adequately.

The buildings and parking areas were kept compact to allow plenty of space for a courtyard, play area, and landscaping. Other community amenities include a small grass lawn; attractively landscaped walkways; and a community room and kitchen, computer lab, and central laundry facility. Each apartment has its own patio or deck. The community is served by public transit and is located within walking distance of local elementary and middle schools, grocery stores,

a low-cost medical clinic, and other neighborhood services.

Compact housing developments can be good for the community and the environment because they typically use land and public infrastructure more efficiently than sprawling developments. But on urban lots it's not easy to create higher-density housing that also allows adequate space for parking, outdoor recreation, and landscaping. For Sara Conner Court Apartments, Eden chose to build many of the homes above a podium parking structure. Although the parking structure costs considerably more than surface parking, it preserves valued open space and contributes to a more livable community.

To establish an attractive and environmentally friendly landscape, the Sara Conner Court team removed topsoil before construction began and stockpiled it away from construction activities. Later the topsoil was respread in the site's planting areas; compost and organic soil amendments were added to supplement missing nutrients; and the planting beds were aerated. Compost adds billions of beneficial organisms to the soil. The compost used at Sara Conner Court is certified by a Seal of Testing Assurance from BFI Organics.

The site is landscaped with plants suited to the Bay Area's Mediterranean climate, which is characterized by a six-month dry season and cool, rainy winters. Selected plants include pineapple guava, barberry, Oregon grape, quince, and rosemary. A key element of the landscaping plan is hydrozoning, which simply means situating plants with similar water needs together so that they're watered by the same valve on the irrigation system. Turf demands a lot of water, so it's particularly important to zone any lawn areas separately from drought-tolerant plants. At Sara Conner Court, the turf is limited to a small lawn in front of the development and another next to the play area, and the turf is irrigated separately with a high-efficiency irrigation system.



As one of its services, Build It Green tracks the number of green homes in different communities.

All rainwater that runs off the parking areas and the roofs drains to the lawns, planting areas, or bioswales. This allows the runoff to percolate into the soil, where organisms can break down contaminants such as motor oil into harmless components. It also recharges the groundwater, reduces erosion, and provides natural irrigation.

A diverse palette of plants creates a beautiful environment for people and attracts birds. Because they are diverse the plants attract fewer pests, which in turn means that pesticide use can be minimized. Most of the plants are also drought tolerant and not invasive, which conserves water and reduces waste.

A Range of Green Services

Besides the GreenPoint Rated System, Build It Green provides a number of other services related to the building industry, for professionals and homeowners alike.

The Ask An Expert service provides customized responses to a variety of green building questions for building professionals and the general public in the nine-county San Francisco Bay Area. Questions are submitted to Build It Green by phone, by e-mail, at an

event, or in person by appointment. This free service addresses every question that is submitted and offers additional suggestions for improving the environmental performance of the project.

Build It Green's Access Green Directory provides an extensive listing of green products and technologies available in the San Francisco Bay

Area. It helps professionals and the general public to locate suppliers and service providers of green building products, ranging from foundation and framing materials to finishing touches.

Build It Green also offers training for professionals involved in the design and construction of residential buildings, as well as professionals who support and develop the market for green building. Opportunities include certified green building professional training, advanced training, Green Points Home Rater training, and various workshops.

The Best Builders program provides building owners and designers with free assistance in the design and construction of their green projects to meet a city's goal of increasing its percentage of green development. The service offers information and guidance on green building strategies, helps to identify and implement green project goals, and provides referrals to resources and incentives.

Build It Green also organizes and sponsors workshops and lunchtime presentations on a variety of green building topics. These events are tailored to specific audiences, consisting of homeowners, building professionals, or public agencies.

Build It Green Home Tours offer a glimpse inside some of California's

greenest homes. These innovative self-guided tours showcase beautiful homes that were built or remodeled using healthy energy- and resource-efficient products and practices.

By providing strategic assistance to local governments, Build It Green helps communities to develop, promote, and implement their green building programs. Build It Green serves as an umbrella and facilitator for associate councils, guilds, and networks throughout California, including the Public Agency Council, the Green Remodelers Guild, the Non-Profit Network, the Green Affordable Housing Coalition, the Real Estate Council, the Suppliers Council, and the Builders Council. These guilds and councils represent key stakeholders needed to transform the building industry in California. Build It Green also maintains a public reference library of books, periodicals, and other source materials and showcases a wide array of environmentally responsible building products, such as permeable concrete, structural insulated panels, Forest Stewardship Council (FSC)-certified wood, and natural linoleum flooring.

With all of these services, it's no wonder that everyone from individual homeowners to large developers has been consulting Build It Green for all of their green building questions. As green building continues to become a more integrated aspect of the home construction business, the services this organization provides can only become more essential across the state of California.



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For more information:

To learn more about Build It Green, go to www.builditgreen.org.